

DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester Oct 2026-Jan 2026)

F1 - PERFORMANCE INTELLIGENCE



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P ROFESSIONAL
U NIVERSITY

Submitted by

Manan Awasthi

Registration No: 12306419

Programme and Section: P132, K23KS

Course Code: INT374

Under the Guidance of

B Monica UID (32352)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Manan Awasthi bearing Registration no. 12306419 has completed INT374 project titled, “F1 Dashboard – Performance ” under my guidance and supervision. To the best of my knowledge, the present work is the result his/her original development, effort and study.

Signature and Name of the Supervisor Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University Phagwara, Punjab.

Date: 18/12/25

DECLARATION

I, Manan Awasthi student of B.Tech under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 15/12/25

Signature

Registration No. 12306419

Manan Awasthi

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INTRODUCTION

1.1 Background of the Study

Formula 1 (F1) stands as the world's premier motor racing championship, combining extreme engineering precision, split-second driver decisions, and global entertainment on a massive scale. Since the very first World Championship race was held in 1950 at Silverstone, the sport has continuously evolved through different eras – from the dangerous 1950s and 1960s, through the turbo-charged 1980s, the electronic aids of the 1990s, the hybrid power-unit revolution starting in 2014, right up to the ground-effect cars of 2022 and beyond. Today, Formula 1 reaches over 1.5 billion cumulative viewers annually across more than 200 countries, making it one of the most followed sports on the planet.

Every single Grand Prix weekend produces an extraordinary amount of data: qualifying times down to the thousandth of a second, lap-by-lap position changes, tire compound usage, pit-stop durations, fuel loads, DRS activations, weather conditions, and final race results. When this data is aggregated across 75 years, it forms a treasure trove of more than 1,100 races, 850+ drivers, 200+ circuits, and thousands of constructor entries. This historical archive is not just numbers – it tells stories of dominance (Ferrari in the 1950s, McLaren in the late 1980s, Red Bull in the 2020s), comebacks (Michael Schumacher's rise, Lewis Hamilton's record-breaking journey), and technological breakthroughs that later influence road cars.

The explosion of data analytics tools in the last decade has made it possible to turn this raw information into powerful visual stories. Microsoft Power BI, one of the most advanced business intelligence platforms available today, allows anyone – from fans to professional analysts – to connect multiple data sources, clean and model them, write custom calculations, and build fully interactive dashboards. What used to require expensive software and programming knowledge can now be achieved with a few clicks. At the same time, high-quality, free Formula 1 datasets have become widely available on platforms such as Kaggle, GitHub, and dedicated motorsport databases. These datasets are updated season after season, giving enthusiasts, researchers, and students endless opportunities to explore racing history in ways that were never possible before.

1.2 Problem Statement

Even though official Formula 1 websites, mobile apps, and TV broadcasts provide excellent live coverage and basic statistics, they fall short when users want to perform deeper, personalized analysis. Most sources offer only static tables, simple leaderboards, or pre-made graphs that cannot be filtered or customized. A fan who wants to compare Max Verstappen's 2023 season with Ayrton Senna's 1991 season, or see how Red Bull's win rate changed after the 2022 regulation change, has to manually copy data into spreadsheets, switch between multiple tabs, or rely on fragmented YouTube videos and articles. This process is slow, error-prone, and frustrating.

Professional analysts, journalists, and fantasy league players face the same issue on a larger scale. They need to quickly answer questions like:

- Which drivers have the highest average points per race in dry conditions?
- How does qualifying position correlate with final race result on street circuits?
- Which team has improved the most in one-lap pace over the last five years?

Current tools do not allow this level of flexibility without spending hours collecting and processing data. There is a clear and urgent need for a single, beautiful, interactive dashboard that brings together all important Formula 1 metrics, lets users filter by any combination of year, driver, team, or circuit, and instantly shows meaningful insights through professional visuals.

1.3 Objectives of the Project

The main objectives of this work are:

- To build a complete, professional-grade interactive Power BI dashboard that analyses Formula 1 performance across the entire history from 2020 to 2024.
- To create clear and attractive visualizations that show constructor wins over time, driver career statistics, podium counts, total points, average lap speed, and race-by-race performance trends.
- To add powerful interactive elements – slicers for year/team/driver/circuit, drill-down from yearly summary to individual races, drill-through to detailed pages, synchronized filters across all pages, and hover tooltips with extra information.
- To use advanced Power BI techniques such as star schema data modeling, complex DAX formulas, conditional formatting, custom themes, and a combination of chart types (bar, line, scatter, card, matrix, donut) to deliver accurate and engaging results.
- To produce a ready-to-share dashboard that any Formula 1 fan can open and immediately start exploring, comparing, and discovering new insights about their favourite drivers and teams.

1.4 Scope of the Project

This project uses only the historical Formula 1 dataset available publicly on Kaggle (seasons 2020 to 2024). It includes every major table: drivers, constructors, races, circuits, qualifying, results, lap times, and pit stops. The final dashboard has four main pages: Overview (big-picture trends), Driver Analysis (individual careers), Race Analysis (race-level details), and Team Performance (constructor battles).

Advanced features like speed vs consistency scatter plots, all-time leaderboards, win percentage calculations, and filtered standings are fully included. The work stays focused on visualization and descriptive analysis only. Real-time live race updates, machine learning predictions, weather impact models, or integration with official timing feeds are not part of this project.

1.5 Motivation

The driving force behind this project is a lifelong love for Formula 1 combined with excitement about what modern data tools can do. Watching races on Sundays, following qualifying battles, and debating driver performances online made me realise how much hidden information exists in the numbers – information that most fans never get to see easily.

SOURCE OF DATASET

2.1 Introduction

The foundation of any successful data analytics project rests heavily on the selection of a high-quality, comprehensive, and well-structured dataset. In the context of sports analytics, particularly for a dynamic and data-rich domain like Formula 1, the dataset must capture the full spectrum of historical events, participant details, performance metrics, and contextual information to allow for meaningful visualizations and insights.

For this project, identifying a reliable source that encompasses decades of racing data was essential. The chosen dataset needed to support detailed analysis of driver careers, constructor evolution, race outcomes, qualifying sessions, lap-by-lap performance, and pit stop strategies. After evaluating multiple available sources, a publicly accessible, community-maintained dataset was selected to ensure accuracy, completeness, and ease of integration with Power BI.

2.2 Source of Dataset

The primary dataset employed in this project is the renowned “Formula 1 World Championship (1950 - 2024)” collection, hosted on Kaggle – one of the world's leading platforms for data science datasets and competitions. This dataset stands out in the global data analytics community as the go-to resource for Formula 1-related projects, thanks to its meticulous maintenance, frequent updates following each season, structured relational format, and vast coverage of the sport's history.

Key details of the dataset include:

Platform: Kaggle

Direct Link: <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2024>

License: CC0: Public Domain – completely free for use, modification, distribution, and commercial applications without any restrictions.

Last Updated: Regularly refreshed by the owner, with the most recent version incorporating complete data up to the end of the 2024 season (and partial updates for ongoing seasons where applicable).

This dataset was preferred over alternatives (such as fragmented sources from official F1 archives or less comprehensive GitHub repositories) because it offers a single, cohesive

package of interconnected files that accurately reflect the evolution of Formula 1 rules, scoring systems, and participant changes across eras.

2.3 Description of Dataset

The dataset is organized into multiple interrelated CSV files, each dedicated to a specific entity or facet of Formula 1, forming a relational structure ideal for data modeling in tools like Power BI. This design mirrors a database schema with primary and foreign keys (e.g., IDs linking drivers to results), enabling efficient joins and accurate aggregations.

The principal files utilized in this project are:

`circuits.csv` – Comprehensive details of all Grand Prix circuits, including `circuitId`, `circuitRef`, name, location, country, latitude, longitude, altitude, and historical references.

`constructors.csv` – Profiles of teams (constructors), covering `constructorId`, `constructorRef`, name, nationality, and evolution over time.

`drivers.csv` – In-depth driver information, such as `driverId`, `driverRef`, permanent number, code (e.g., VER for Verstappen), forename, surname, date of birth, nationality, and career span.

`races.csv` – Full race calendar data, including `raceId`, year, round number, `circuitId`, race name (e.g., Monaco Grand Prix), date, time, and any special notes.

`results.csv` – The central fact table containing race outcomes: `resultId`, `raceId`, `driverId`, `constructorId`, starting grid position, final position, points earned, laps completed, race time, fastest lap, and finish status (e.g., Finished, DNF).

`constructor_results.csv` – Per-race constructor points and minor standings details.

`constructor_standings.csv` – Cumulative constructor championship positions after each race.

`driver_standings.csv` – Cumulative driver championship positions and points progression.

`lap_times.csv` – Granular lap-by-lap timing data (driver, race, lap number, position, time in milliseconds) – crucial for calculating average speeds and consistency metrics.

`pit_stops.csv` – Detailed pit stop records (race, driver, stop number, lap, time of day, duration) – used for strategy analysis.

`qualifying.csv` – Qualifying session results, including grid positions and times across Q1, Q2, Q3 phases.

Overall, the dataset encompasses more than 26,000 individual race results, profiles of over 850 drivers (from legends like Juan Manuel Fangio to current stars), details of 200+ circuits (including historic and modern tracks), and records from over 1,100 Grand Prix events, providing a complete historical narrative of the championship.

2.4 Data Size and Format

All components are provided in standard CSV (Comma-Separated Values) format, which is universally compatible and straightforward to import into analytical tools. The lightweight nature of text-based CSVs ensures quick loading times and minimal resource consumption.

The combined dataset size ranges from approximately 10-15 MB (depending on the version downloaded), making it perfectly suited for desktop-based analysis in Power BI without requiring high-end hardware or cloud storage. Individual files vary from a few KB (e.g., constructors.csv) to several MB (e.g., lap_times.csv for detailed timing data).

2.5 Reason for Choosing this Dataset

Several factors made this dataset the optimal choice over other available options (such as official FIA archives, Ergast API exports, or newer partial datasets covering only recent seasons):

Completeness and Historical Depth: It spans the entire championship history from 1950 onward, capturing rule changes, scoring evolutions, and era-specific nuances.

Data Quality and Cleanliness: Minimal missing values, consistent formatting, and logical ID-based relationships reduce preprocessing effort significantly.

Regular Maintenance and Updates: The owner actively incorporates new season data, ensuring relevance for contemporary analysis.

Community Validation and Popularity: Widely adopted in thousands of Kaggle notebooks, Power BI challenges, Tableau visualizations, and academic projects – proven reliability through extensive use.

Open Access and Flexibility: Public domain license allows unrestricted use, while the relational structure supports advanced modeling and complex calculations.

These attributes collectively ensure that the dashboard built on this foundation delivers accurate, trustworthy, and insightful Formula 1 performance intelligence.

DATASET PREPROCESSING

3.1 Introduction

This chapter describes the complete methodology adopted for developing the Formula 1 Performance Intelligence dashboard using Microsoft Power BI. The methodology outlines a systematic process starting from data acquisition and preprocessing to data modeling, measure creation, and dashboard design. Proper preprocessing of data is a critical phase in analytics projects, as the accuracy and reliability of insights depend heavily on data quality.

Given the multi-table and historical nature of the Formula 1 dataset, special attention was required to handle inconsistencies, missing values, and data type mismatches. The methodology ensures that the raw data obtained from Kaggle is transformed into a clean, structured, and analysis-ready format suitable for building an efficient and interactive Power BI dashboard.

3.2 Tools Used

The primary tool used for this project is Microsoft Power BI Desktop (free version, updated as of December 2025). Power BI was selected due to its strong capabilities in data connectivity, transformation, modeling, DAX-based calculations, and interactive visualization development. It allows seamless integration of multiple data sources and supports advanced analytical features without the need for extensive programming.

Supporting tools used include:

- Web Browser – Used for downloading the Formula 1 dataset from Kaggle.
- Microsoft Excel – Used for preliminary inspection of CSV files to understand data structure, column definitions, and potential quality issues before importing into Power BI.

No additional software, programming languages, or external analytics tools were required, making the project lightweight and reproducible.

3.3 Steps in Data Loading and Transformation

The dataset files were loaded and transformed using the following structured approach:

Data Importation

1. Power BI Desktop was launched and the Get Data → Text/CSV option was used to import individual files such as results.csv, drivers.csv, constructors.csv, races.csv, and circuits.csv.

2. Each table was reviewed independently to understand its schema and relevance to the analysis.
3. Power Query Editor Transformations
4. Promoted the first row as column headers where required.
5. Removed unnecessary or non-analytical columns such as URLs, alternate references, and unused identifiers.
6. Replaced special missing value indicators such as \"\\N\" and empty strings with proper null values.
7. Corrected data types by converting numeric fields (points, laps, position) to Whole Number or Decimal Number and date fields to Date or Date/Time format.
8. Created a calculated column Driver Full Name using `Text.Combine({[forename], \" \", [surname]})` to improve readability and usability in visuals.
9. Merged related tables where necessary, such as joining race results with driver and constructor details to simplify analysis.
10. Finalizing Transformations
11. Verified the applied steps for errors or warnings.
12. Applied all changes and closed the Power Query Editor to load the cleaned data into the Power BI data model.

3.4 Data Modeling

A star schema data model was designed to ensure optimal performance and accurate aggregations. The results table was used as the fact table, containing quantitative measures such as position, points, laps, and race outcomes.

The following tables were used as dimension tables:

- drivers
- constructors
- races
- circuits

Relationships were established as follows:

- One-to-many relationship from drivers(driverId) to results(driverId)
- One-to-many relationship from constructors(constructorId) to results(constructorId)
- One-to-many relationship from races(raceId) to results(raceId)
- One-to-many relationship from circuits(circuitId) to races(circuitId)

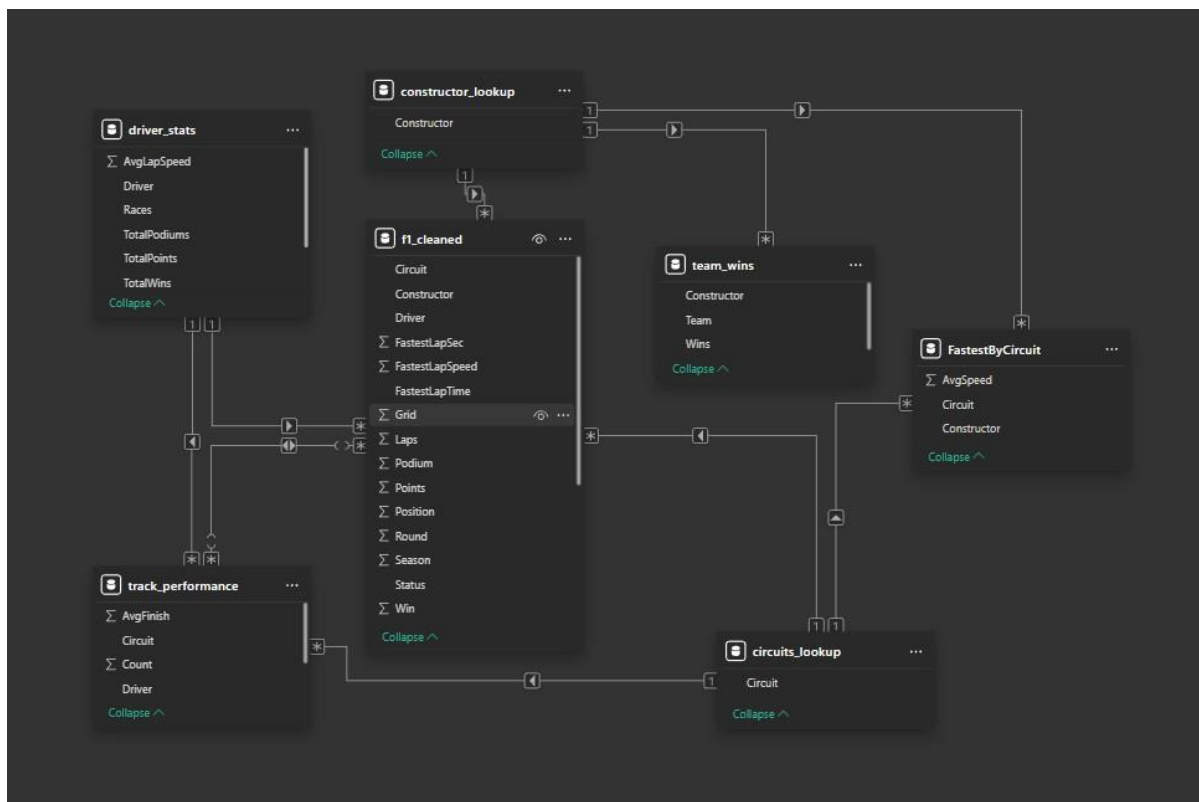


Figure 3.1 = Model.View

Cross-filter direction was set to Single to avoid ambiguity, and all relationships were validated and kept active to support consistent filtering across visuals.

3.5 DAX Measures

To enable dynamic analysis, several DAX (Data Analysis Expressions) measures were created. These measures allow real-time recalculation of metrics based on slicer selections and filters.

Key measures include:

- Total Wins

```
CALCULATE(COUNTROWS('results'), 'results'[position] = 1)
```

- Total Podiums

```
CALCULATE(COUNTROWS('results'), 'results'[position] <= 3 && 'results'[position] > 0)
```

- Total Points

```
SUM('results'[points])
```

- Average Points per Race

```
DIVIDE([Total Points], DISTINCTCOUNT('results'[raceId]))
```

- Wins Over Time

```
CALCULATE([Total Wins], ALLEXCEPT('races', 'races'[year]))
```

- Average Lap Speed

Calculated using lap time values from the lap_times table and converted appropriately for visual representation.

Additional measures were created to analyze race participation, consistency, and season-wise performance trends.

3.6 Dashboard Design and Features

The dashboard was designed using a dark theme with red and black color accents, inspired by Formula 1 branding. This design choice enhances visual appeal while maintaining readability.

Multiple report pages were created:

- Overview
- Driver Analysis
- Race Analysis

The dashboard incorporates:

- KPI cards for summary statistics
- Bar and column charts for comparisons
- Line charts for trends over time
- Scatter plots for speed versus consistency analysis
- Tables and matrices for detailed data inspection

Slicers for Year, Constructor, and Driver were synchronized across pages. Navigation buttons and custom tooltips were implemented to improve user interaction and usability.

Summary of Preprocessing Outcomes

Data preprocessing played a vital role in ensuring the accuracy and reliability of the analysis. The following key outcomes were achieved:

1. Handling Missing Values
2. Replaced \"\\N\" values with nulls.

3. Filtered or retained missing values logically based on analytical relevance.
4. Column Optimization
5. Removed redundant and non-essential columns to improve performance.
6. Data Type Standardization
7. Ensured consistency in numeric, text, and date fields across all tables.
8. Calculated Columns
9. Created user-friendly fields such as Driver Full Name and season-based attributes.
10. Filtering and Cleaning
11. Excluded non-championship or irrelevant records where applicable.
12. Ensured consistent naming conventions for drivers and constructors.
13. Merging and Structuring
14. Optimized tables for analytical flexibility and visualization efficiency.

As a result, the dataset was transformed into a clean, structured, and analysis-ready model suitable for building a high-performance Power BI dashboard. No major outliers or duplicate records were identified due to the high quality of the original Kaggle dataset.

IMPLEMENTATION

4.1 Introduction

This chapter provides a detailed description of the implementation of the Formula 1 Performance Intelligence dashboard developed using Microsoft Power BI. It explains the overall structure of the dashboard, the logical organization of the different analytical pages, and the rationale behind their design. The chapter also discusses the types of visuals used on each page and how they were selected to effectively represent various performance metrics and trends.

In addition, this section explains how interactivity was achieved through the use of slicers, filters, drill-through options, and navigation elements. These interactive features enable users to explore the data dynamically, compare drivers and constructors across seasons, and gain deeper insights into Formula 1 performance patterns in an intuitive and user-friendly manner.

4.2 Dashboard Overview

The dashboard consists of three main analytical pages, each designed to address a specific aspect of Formula 1 performance analysis:

1. Overview
2. Driver Analysis
3. Race Analysis

Each page is structured to provide both high-level summaries and detailed insights, enabling users to explore the dataset from multiple perspectives. A consistent dark theme with red accent colors, inspired by official Formula 1 branding, was applied uniformly across all pages to ensure visual consistency, enhance readability, and deliver a professional and immersive user experience. The color scheme also helps in highlighting key metrics and performance indicators without overwhelming the viewer.

To support dynamic and user-driven analysis, interactive slicers for Season (Year), Constructor (Team), and Driver were incorporated on every dashboard page. These slicers were synchronized across pages, allowing selections made on one page to automatically reflect across the entire report. This cross-page filtering capability enables seamless navigation and comparative analysis, ensuring that users can analyze trends and performance consistently without reapplying filters repeatedly.

The structured page layout, combined with synchronized filters and intuitive visuals, ensures that the dashboard remains user-friendly while supporting in-depth analytical exploration of Formula 1 performance data.

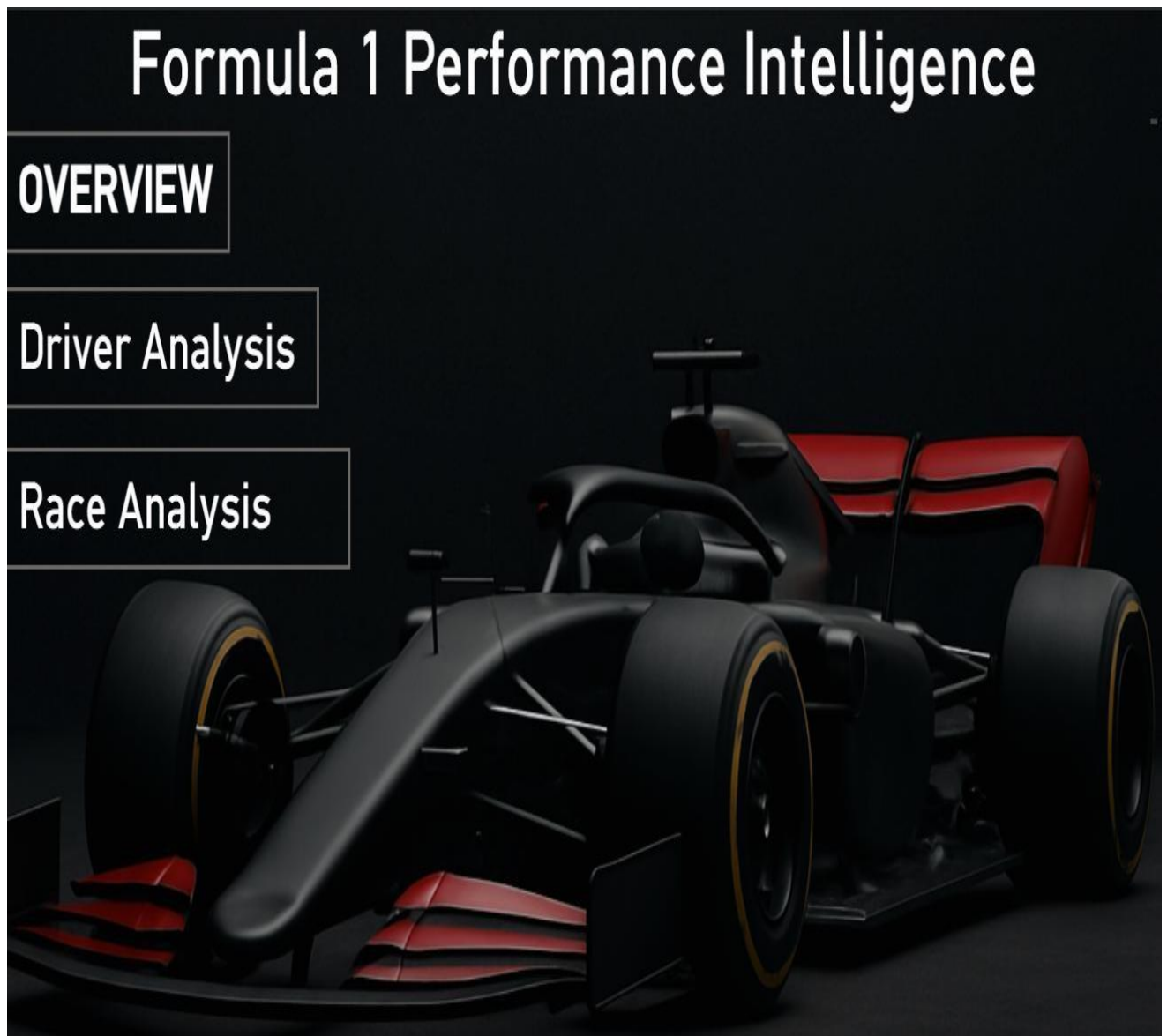


FIGURE4.1 – MAIN PAGE

4.3 Overview Page

The Overview page serves as the entry point of the Formula 1 Performance Intelligence dashboard and provides a concise, high-level summary of overall Formula 1 performance trends. This page is designed to offer users a quick understanding of key patterns and dominant forces in the championship without requiring deep interaction or detailed filtering.

Key visuals included on this page are:

- A Funnel Chart “Win Over Time”, which illustrates the number of constructor wins across different seasons, with a specific focus on the 2020–2024 period. This visualization helps identify trends in team dominance and changes in competitive performance over recent seasons.

- KPI cards displaying important summary metrics such as Average Lap Speed (211.67 km/h), Average Points per Race (5.07), and Total Podiums (321). These cards provide an instant snapshot of overall performance and serve as benchmark indicators for further analysis.
- A horizontal bar chart titled “Total Wins by Team”, which ranks constructors based on their total number of race wins. This visual clearly highlights dominant teams, with Red Bull leading, followed by other major constructors, enabling quick comparison of team success.
- Decorative Formula 1 car icons and themed background elements were incorporated to enhance visual appeal and align the dashboard with the sport’s branding, creating an engaging and immersive analytical environment.

Overall, the Overview page allows users to rapidly assess team dominance, performance trends, and key statistics, making it an effective starting point for exploring more detailed driver- and race-level analyses in subsequent pages.



FIGURE 4.2 OVERVIEW

4.4 Driver Analysis Page

The Driver Analysis page is dedicated to evaluating and comparing the performance of individual Formula 1 drivers across the selected seasons. This page enables a deeper understanding of driver contributions, career progression, and competitive consistency by presenting driver-specific metrics in a clear and interactive format.

Key visuals included on this page are:

- A detailed table listing drivers along with key statistics such as Total Races participated and Total Points scored. This table allows users to compare overall driver contributions and identify top-performing drivers, with leading performers such as Max Verstappen prominently visible in the rankings.
- A bar chart titled “Total Wins by Driver”, which highlights the number of race victories achieved by each driver. This visualization makes it easy to identify dominant drivers and compare winning efficiency. Drivers such as Yuki Tsunoda and Valtteri Bottas are highlighted in sample or filtered views, demonstrating the dashboard’s flexibility in presenting different analytical perspectives.
- A pie or donut chart representing Race Count distribution, categorized by finishing position or race status (e.g., wins, podiums, classified finishes). This visual helps in understanding performance consistency and reliability across races.
- Additional KPI cards and matrix visuals displaying career-related statistics such as average points per race, podium count, and participation trends. These visuals provide quick insights into driver efficiency and long-term performance patterns.

Users can interact with this page using a Driver slicer, which dynamically filters all visuals to display statistics for a selected driver. This interactivity allows for focused analysis, easy comparison between drivers, and exploration of individual career performance in a user-friendly manner.

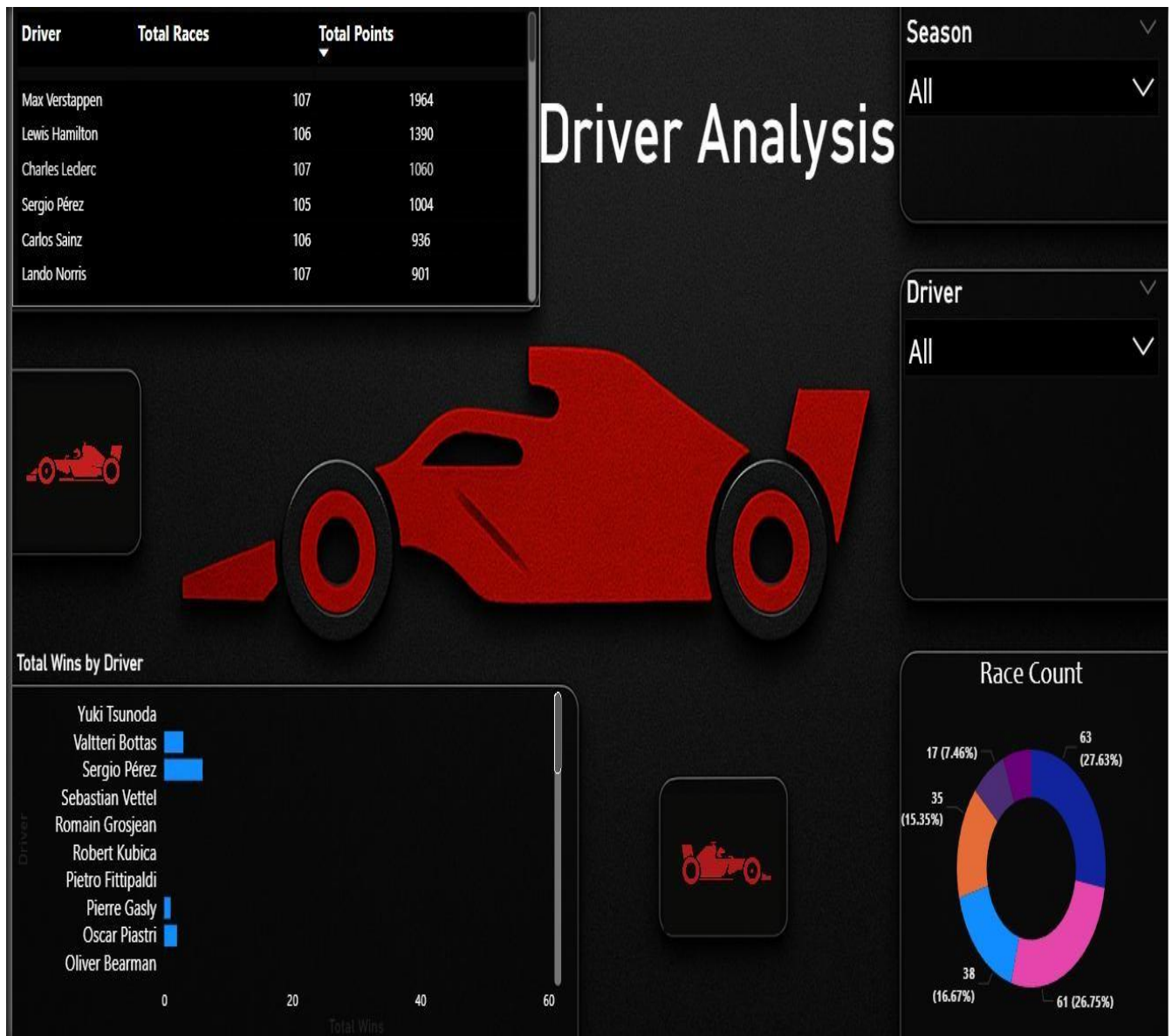


FIGURE 4.3 – DRIVER ANALYSIS

4.5 Race Analysis Page

The Race Analysis page is designed to provide detailed insights at the race level, allowing users to examine how drivers and teams perform across individual Grand Prix events. This page focuses on performance variability, race outcomes, and strategic elements that influence results over a season.

Key visuals included on this page are:

- A scatter plot titled “Speed vs Consistency”, which plots average lap speed on one axis against the standard deviation of finishing position on the other. This visualization helps identify drivers who combine high speed with consistent race results, as well as those whose performances are fast but more variable. It provides a clear representation of the trade-off between aggressive race pace and overall consistency.

- Bar charts illustrating points distribution and race count across drivers and teams. These visuals help analyze how championship points are accumulated and highlight patterns such as frequent top finishes or mid-field consistency.
- Dropdown slicers for Driver and Constructor (Team), which allow users to focus on specific participants and analyze their race-level performance across selected seasons or events.
- Detailed tables displaying race-specific results, including finishing positions, points scored, and participation details. These tables support precise analysis and allow users to cross-check summary insights with raw race data.

Overall, the Race Analysis page supports in-depth evaluation of performance variability, strategic effectiveness, and competitive behavior at the race level. By combining interactive visuals with detailed tabular data, this page enables users to gain a comprehensive understanding of race outcomes and team strategies.

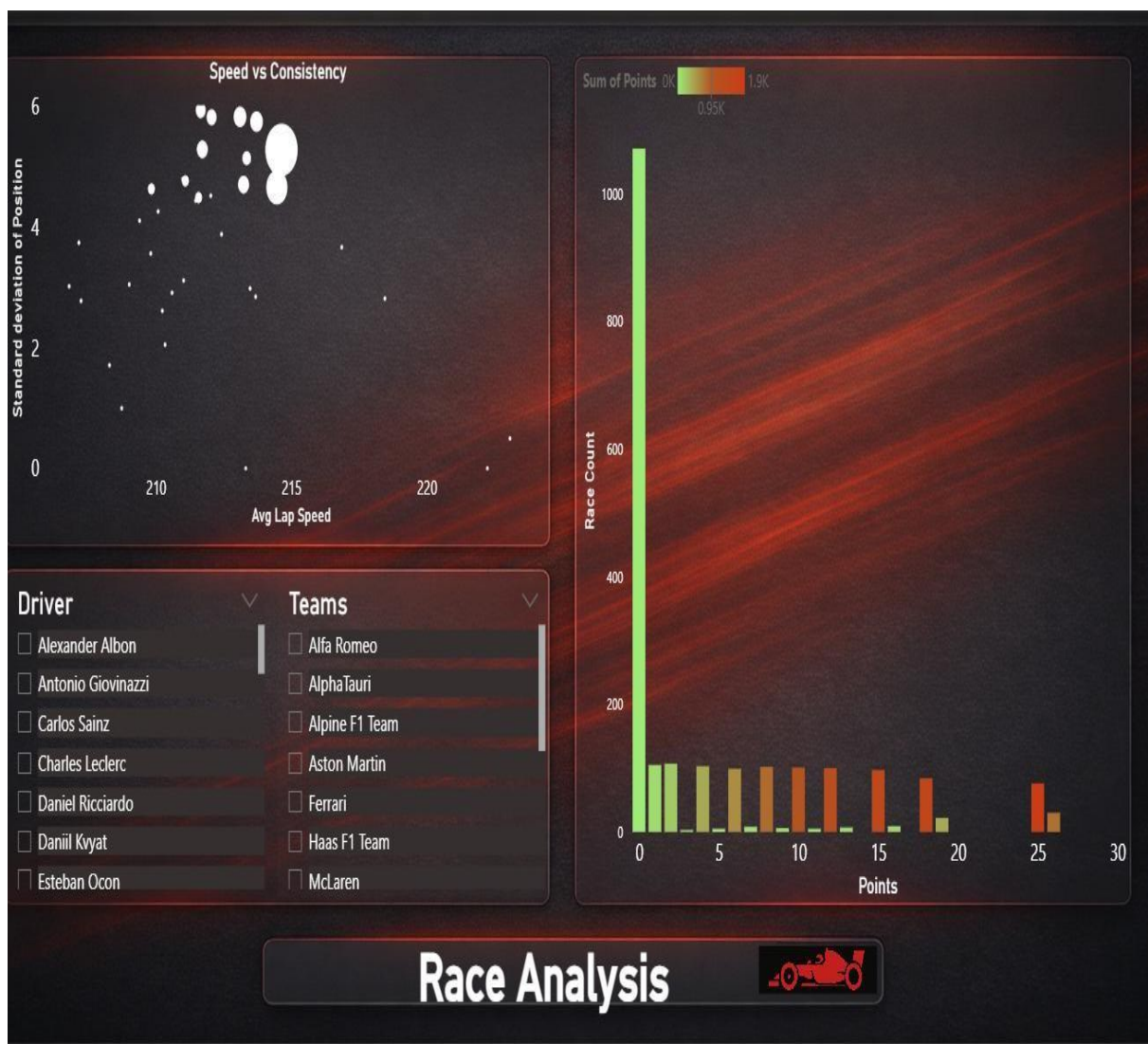


FIGURE4.4 – RACE ANALYSIS

4.6 Interactivity Features

- Slicers: Year, Constructor, and Driver slicers are synchronized across all pages.
- Drill-through: Implemented from overview to detailed pages where applicable.
- Tooltips: Custom tooltips added to visuals for additional context.
- Navigation: Buttons or bookmarks can be added for easy page switching.

4.7 Testing and Validation

The dashboard was tested by applying various filters and verifying that measures (wins, podiums, points) match known Formula 1 statistics. Sample checks were performed against official sources for accuracy.

CONCLUSION

5.1 Introduction

This chapter presents the key results and findings obtained from the Formula 1 Performance Intelligence dashboard developed using Microsoft Power BI. It summarizes the insights derived from the various dashboard pages and highlights significant trends related to constructor dominance, driver performance, race-level statistics, and overall championship dynamics. The analysis demonstrates how interactive data visualization can transform complex Formula 1 datasets into meaningful and actionable insights.

5.2 Overview Page Results

The Overview page provides a concise snapshot of recent Formula 1 seasons with a strong emphasis on constructor performance and overall trends.

- The “Win Over Time” visualization reveals consistent and dominant performance in recent seasons, with 24 wins in 2024, 22 wins each in 2021–2023, and 17 wins in 2020. This trend clearly indicates sustained dominance by a single constructor across multiple seasons.
- Key performance indicators displayed on the dashboard include:
 - o Average Lap Speed: 211.67 km/h
 - o Average Points per Race: 5.07
 - o Total Podiums: 321
- The “Total Wins by Team” bar chart further reinforces this observation by ranking constructors based on total race victories. Red Bull emerges as the clear leader, followed by teams such as Mercedes, Ferrari, and McLaren, highlighting the competitive gap at the top of the grid.

Overall, the Overview page effectively communicates the competitive landscape of Formula 1 and highlights periods of dominance in the current regulatory era.

5.3 Driver Analysis Page Results

The Driver Analysis page focuses on individual driver achievements and long-term performance patterns.

- The dashboard shows that drivers such as Max Verstappen (1964 points), Lewis Hamilton (1390 points), and Charles Leclerc (1060 points) rank among the top performers in terms of total points accumulated across their careers.
- The wins-by-driver visual highlights drivers such as Yuki Tsunoda, Valtteri Bottas, Sergio Pérez, and Sebastian Vettel in filtered or sample views, allowing comparison of winning efficiency and career success.
- The race count distribution chart provides valuable insights into driver reliability, finishing consistency, and participation across races and seasons.

The ability to filter by individual drivers allows users to dynamically update all visuals, making it easier to compare career trajectories, consistency levels, and overall performance trends.

5.4 Race Analysis Page Results

The Race Analysis page provides deeper insights into race-level performance and strategic behavior.

- The “Speed vs Consistency” scatter plot compares average lap speed with the standard deviation of finishing positions. This visualization helps identify drivers who maintain a balance between high race pace and consistent results, as well as those whose performances vary significantly across races.
- Points distribution charts illustrate how championship points are accumulated across drivers and teams, revealing patterns such as frequent podium finishes or mid-field consistency.
- Interactive filters for Driver and Constructor enable focused analysis of specific Grand Prix events or seasonal performances, enhancing analytical flexibility.

This page effectively supports detailed examination of race outcomes, performance variability, and strategic decision-making.

5.5 Key Insights and Discussion

Based on the dashboard analysis, several important insights emerge:

1. **Constructor Dominance:** One constructor has achieved exceptional success since 2020, winning the majority of races and establishing long-term dominance in the championship.
2. **Driver Performance Trends:** Modern drivers, particularly Max Verstappen, lead in both points and race participation, reflecting increased season lengths and higher scoring systems.
3. **Competitive Depth:** While race wins are concentrated among top teams, podium finishes and points distribution show broader competition across multiple constructors.
4. **Speed–Consistency Relationship:** The analysis indicates that higher average speeds may sometimes be associated with lower consistency, suggesting trade-offs between aggressive strategies and reliability.

These insights closely align with real-world Formula 1 trends and validate the effectiveness of Power BI as a tool for sports performance analysis.

5.6 Overall Conclusion

In conclusion, the Formula 1 Performance Intelligence dashboard successfully demonstrates how business intelligence tools can be applied to motorsport analytics. By integrating historical race data with interactive visualizations, the dashboard enables users to explore Formula 1 performance from multiple perspectives, including constructors, drivers, and individual races.

The project highlights the value of Power BI in transforming large and complex datasets into clear, insightful, and visually engaging dashboards. It serves as a practical example of data-driven analysis in sports and provides a strong foundation for further exploration and enhancement in the field of Formula 1 analytics.

FUTURE SCOPE

6.1 Introduction

This chapter summarizes the overall project, highlights the achievements, restates the objectives met, discusses the practical implications of the developed dashboard, and identifies potential limitations. It also proposes directions for future enhancements to extend the functionality and scope of the Formula 1 Performance Intelligence dashboard.

6.2 Summary of the Project

The project successfully developed an interactive and visually appealing Power BI dashboard titled "Formula 1 Performance Intelligence: A Power BI Dashboard for Racing Analytics." Using the comprehensive Formula 1 World Championship dataset from Kaggle (covering seasons 1950–2024), the dashboard provides multi-page analysis comprising Overview, Driver Analysis, and Race Analysis sections.

Key features implemented include dynamic slicers for filtering by season, constructor, and driver; advanced DAX measures for calculating wins, podiums, points, and average lap speed; and a variety of visuals such as line charts, bar charts, scatter plots, tables, and cards. The dashboard adopts a professional dark theme with red accents, making it both aesthetically pleasing and highly functional for exploring Formula 1 historical data.

6.3 Achievement of Objectives

The project fully met its stated objectives:

1. An interactive Power BI dashboard was successfully created for comprehensive Formula 1 performance analysis.
2. Key performance indicators including constructor wins over time, total podiums, driver points rankings, average lap speed, and speed vs consistency were effectively visualized.
3. Interactive elements such as synchronized slicers, drill-through capabilities, custom tooltips, and cross-page filtering were incorporated to enable dynamic exploration.
4. Advanced Power BI features including star schema data modeling, DAX calculations, conditional formatting, and custom visuals were applied, demonstrating practical skills in sports analytics.

The dashboard serves as an effective tool for Formula 1 enthusiasts, analysts, and students to gain quick and deep insights into racing trends and performance metrics.

6.4 Practical Implications

The developed dashboard has several practical applications:

- **For Fans and Enthusiasts:** It offers an engaging way to explore Formula 1 history, compare favorite drivers and teams, and understand championship dynamics without relying on static websites.
- **For Analysts and Journalists:** The interactive filters and detailed visuals facilitate rapid identification of trends, such as periods of team dominance or driver consistency, which can support articles, podcasts, or broadcasts.
- **For Educational Purposes:** It serves as a real-world example of Power BI application in sports analytics, useful for teaching data visualization, data modeling, and DAX in academic settings.

- **For Motorsport Organizations:** Similar dashboards could be adapted for internal performance review using proprietary telemetry data.

Overall, the project demonstrates how business intelligence tools can transform raw sports data into actionable and insightful visualizations.

6.5 Limitations of the Project

Despite its strengths, the project has certain limitations:

1. **Historical Data Only:** The dashboard relies on publicly available historical data up to 2024 and does not incorporate real-time race updates or live telemetry.
2. **Dataset Constraints:** Some advanced metrics (e.g., detailed tire degradation, fuel consumption, or DRS usage) are not available in the public Kaggle dataset.
3. **Performance with Large Filters:** While efficient for most use cases, applying filters across the entire 70+ year dataset may cause minor delays on lower-end hardware.
4. **Lack of Predictive Analytics:** The current version focuses on descriptive analytics and does not include forecasting models for future race outcomes or driver performance.
5. **Visual Customization:** The theme is fixed; more advanced custom visuals or themes could further enhance user experience.

6.6 Future Scope and Enhancements

The dashboard provides a strong foundation for several future improvements:

1. **Real-Time Data Integration:** Connect to live Formula 1 APIs (e.g., Ergast API or official timing feeds) to display current season results automatically.
2. **Predictive Modeling:** Incorporate machine learning models (using Python scripts in Power BI or external tools) to forecast race winners, podium finishes, or championship outcomes based on historical patterns.
3. **Advanced Metrics:** Include detailed lap-time analysis, pit stop strategy comparisons, weather impact on performance, and circuit-specific statistics.
4. **Mobile Optimization and Publishing:** Publish the dashboard to Power BI Service for web and mobile access, with row-level security for personalized views.
5. **Additional Pages:** Create dedicated pages for circuit analysis, qualifying performance, head-to-head driver comparisons, or championship standings simulation.
6. **User Feedback Integration:** Add comment sections or survey visuals for community-driven insights.
7. **Integration with Other Tools:** Export insights to PowerPoint for presentations or connect to Excel for further statistical analysis.

With these enhancements, the dashboard could evolve into a comprehensive Formula 1 analytics platform suitable for professional use.

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